

Stress-Relief Re-Analysis — Slide Content

Audience: technical peers / design review **Format:** draft content for your own template

Each slide below gives **on-slide bullets** (tight, what the audience reads), **picture** (what to show and how to capture it), and **talking points** (what you say, written to pre-empt the questions a reviewer will actually ask).

Slide 1 — Introduction / Why we're here

On-slide bullets

- Dome geometry was revised; the stress-relief (PWHT cooldown) report was not re-run against it.
- Objective: update the analysis to the new dome **without rebuilding the model**.
- Approach: keep the original, validated analysis step intact; **stitch only the changed geometry** into the existing orphan mesh via a tie.
- Everything else — loads, BCs, cavity radiation, material, step definition — is reused unchanged.

Picture

- Side-by-side: original dome vs revised dome, with the changed region called out (a box or color highlight on the region that actually moved). If you have both CAD states, a simple overlay with the delta shaded is the strongest single image.
- Optional small inset: the three acceptance criteria (25 °F/hr min cooling rate, 100 °F max ΔT on a weldment, stress $\leq 1.3 \cdot S_y$) so the room knows what "the analysis" is verifying.

Talking points

- The report is a qualification artifact — it has to reflect as-built geometry. A geometry change with a stale analysis is an open finding, not a paperwork item.
 - The expensive part of this model is the cavity-radiation view-factor solve and the validated step setup. Rebuilding from scratch would mean re-validating from scratch. The whole point of the tie approach is to **change the one thing that changed and inherit the rest** — including the validation basis.
 - Frame the deliverable honestly: this is a *targeted re-qualification*, not a new analysis. The bar is "prove we changed only what we intended, and the criteria still hold."
 - Anticipated question — "*why not just remesh the whole thing?*" Answer: a full remesh breaks node-to-node temperature transfer into the structural pass and forces re-validation of the entire deck. The tie keeps the untouched mesh bit-identical, so the far field is a built-in control (Slide 4).
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Slide 2 — What it took: the region swap and change control

On-slide bullets

- New dome region meshed independently, renumbered to **20001+** so labels can't collide with the original mesh.
- New (denser) region tied in as the **secondary** surface; original mesh is the **main** surface.
- Deleting the old region and adding the new one breaks references **silently** — the job still runs.
- Tracked and re-established across every affected entity:
 - element / node **labels** (deleted vs added)
 - **material** + section assignment (incl. cylindrical orientation)
 - **loads** / **BCs** (prescribed-temperature sets)
 - **interactions** — cavity radiation surfaces & view factors
 - **sets & surfaces** feeding all of the above

Picture

- The tied interface: original orphan mesh in one color, new 20001+ region in another, tie interface highlighted. A zoom on the seam showing the density mismatch (fine secondary against coarser main) makes the "why secondary" point visually.
- Optional: a small slice of the change-control ledger (a few rows) so the room sees this was enumerated, not eyeballed.

Talking points

- Lead with the risk, because it's the interesting part: in a deck with **hundreds of sets and surfaces wired into 15+ cavity enclosures**, deleting a region doesn't throw an error. The set quietly loses members, the surface quietly loses faces, the job runs, and it hands you a plausible, wrong field. That's the failure mode this whole slide exists to defeat.
- Three concrete silent breaks worth naming: a temperature-BC set that loses its nodes → that region goes **unconstrained**; a cavity surface that loses faces → **stale view factors**, radiating area wrong; the new region with **no section assigned** → drops out of the answer. None of these warn you.
- Why the new region is **secondary**: it's the finer mesh, and the finer side should always be secondary so its many nodes are each tied to the coarser main surface — smoothest load/flux transfer. Denser-as-main gives a lumpy interface.
- Why **20001+**: guarantees globally unique labels so nothing — sets, surfaces, or a label-keyed subroutine — can accidentally address an old element that no longer exists or a new one it doesn't know about.

- The message for a reviewer: this wasn't "delete and hope." It was formal change control — enumerate every impacted entity, act, verify, and the outstanding-item count had to read zero before submitting.
 - Anticipated question — *"how do you know you caught everything?"* Answer: the impacted entities were pulled programmatically from the deleted-label lists, not hand-scanned, and the recompute was verified (VFTOT $\approx$ 1.0 on rebuilt cavities, no unassigned-section / empty-set warnings at datacheck).
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Slide 3 — Temperature is continuous across the tie

On-slide bullets

- Thermal tie enforces **temperature continuity** across the new-to-original seam.
- Contour/animation shows **no kink, no jump** in NT across the interface through the full cooldown.
- Confirms the tie is transferring the field correctly despite the mesh-density mismatch.

Picture

- Animation of the NT contour over the cooldown transient, camera framed on the seam. What the audience should see: the color band crosses the tie line **smoothly** — no discontinuity, no isolated hot/cold ring at the interface.
- If you can only show a still: pick the instant of maximum through-wall gradient (worst case for revealing a bad tie), not $t=0$.

Talking points

- This is the **primary validation of the tie itself**. A tie between dissimilar mesh densities is where you'd expect trouble; a smooth field across it is direct evidence the constraint is well-formed and the position tolerance captured the intended faces.
 - Continuity here is also the precondition for the structural pass. The temperature field crossing this seam is what gets mapped into the stress model — a kink in NT at the seam would propagate straight into a spurious stress concentration. Smooth here means clean there.
 - Point out *what a failure would have looked like* so the room can see you checked for it: a visible temperature step at the interface, or a localized artifact ring, would mean secondary nodes weren't tied or were tied across a gap. We don't see that.
 - Anticipated question — *"is that smoothness real or just contour interpolation smoothing over a problem?"* Answer: cross-check is the far-field match on Slide 4 and the VFTOT audit — the smooth contour isn't the only evidence, it's corroborated.
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Slide 4 — Identical-node temperature comparison: modified vs. validation

On-slide bullets

- Overlay: T-vs-time at **identical node locations**, modified model vs. original (validation) analysis.
- **Far-field nodes** (untouched orphan mesh): essentially **identical** to validation → confirms we changed only what we intended.
- **Modified dome region**: cools **faster** than the original dome.
- Cooling-rate and ΔT criteria re-checked against the new histories.

Picture

- Two-panel plot, or one plot with grouped curves:
 - **Panel A — far field**: modified and validation curves overplotted; they should sit on top of each other. This is the control.
 - **Panel B — modified dome**: modified vs. original dome curves separating, modified below.
- Use the *same node IDs* in both models (the untouched region has identical labels; label the modified-region nodes by physical location). Mark the governing-criteria times if they're visible.

Talking points

- Lead with the **far-field match — it's the headline result**. Untouched nodes reproducing the validation analysis is the proof that the region swap was surgical: no unintended change leaked into the rest of the model. That single overlay retires most of "did you break something" for a reviewer.
- On "**the modified dome cools faster**" — **tighten the mechanism before you present it**. The bullet you gave me said "*more material for heat transfer*," and a reviewer will stop you on it: **more material means more thermal mass, which cools *slower*, not faster**. More mass is the opposite of what you're claiming. So the real driver is almost certainly one of:
 - **more radiating / surface area** participating in the cavity (more area facing the sink at fixed mass → faster);
 - a **thinner or redistributed wall** in the new dome → shorter conduction path, less through-wall lag;
 - **changed view factors** exposing more of the region to the cold boundary.

Any of those is defensible; "*more material*" is not, as phrased. Pull the actual geometry delta (Δ surface area, Δ wall thickness, Δ mass) and state the specific one. If mass genuinely went **up** and it still cools faster, then surface-area or view-factor gain is dominating the mass increase, and that's a more interesting — and more precise — story to tell. **Recommend confirming this from the geometry before the review**, because it's the one claim on these four slides most likely to draw fire.

- Faster cooling in the modified region is **not automatically benign**: faster local cooling can raise the local cooling rate and the local ΔT . Close the loop explicitly — state that the new histories were re-checked against the 25 °F/hr floor and the 100 °F ΔT limit, and that they still pass (or note the margin if it moved).
 - Anticipated question — *"different mesh density on the two sides — is the comparison apples-to-apples?"* Answer: far-field nodes are label-identical between the two models, so that comparison is exact; the modified-region comparison is location-based by necessity, which is the honest caveat.
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Slide 5 (optional) — Result & confidence

Recommended closer; Slide 4 ends on evidence, this ends on the conclusion.

On-slide bullets

- Analysis updated to the revised dome by targeted region swap; validated step setup preserved.
- Far-field temperature reproduces the original analysis → change was contained.
- Acceptance criteria (25 °F/hr, 100 °F ΔT , 1.3·Sy) re-verified on the new geometry: **[state pass / margins]**.
- Change control closed: all impacted sets, surfaces, cavities, loads, and sections re-established and verified.

Picture

- A compact criteria table: each criterion, original result, modified result, margin. A design-review audience reads a results table faster than prose.
- Optional: the worst-case ΔT contour on the modified weldment at the governing instant.

Talking points

- The confidence argument rests on **three independent checks agreeing**: smooth NT across the tie (Slide 3), far-field reproduction of validation (Slide 4), and a clean change-control close-out (Slide 2). Different failure modes, same conclusion.
 - Be explicit about scope: this re-qualifies the **thermal** cooldown for the new dome. If the structural pass is downstream, say so and note it inherits this temperature field.
 - Close on the open items honestly — anything still to confirm (e.g. the exact cooling-rate margin in the fastest-cooling new segment) is more credible flagged by you than found by the panel.
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One cross-cutting note

The strongest through-line for this deck is "**we changed one thing and proved the rest is untouched.**" Slides 2-4 are three angles on that same claim (enumerated change control, continuous seam, far-field match). If you're tight on time, that's the spine — and the far-field overlay on Slide 4 is the single most persuasive image in the set. Lead your verbal summary with it.

The one content fix to make before the review is the Slide 4 mechanism wording — "*more material*" → the specific geometric driver. It's the only claim here that's likely wrong as stated.